iLLD-TC4-v2.4.1

Release Notes

Product name: TC4xx_SW_iLLD
Release number and version: iLLD-TC4-v2.4.1
Type of release: Patch *
Release method: Github
Processor platform: TC4Dx AA, TC4Dx AB
Date: 2025-11-14
Previous release number and version: Not Applicable

About this document

Scope and purpose

This document details the release contents, all known issues in the release, and tool information.

*Note: * This release is not intended for production use. This is reference software that helps the customers to develop their software.*

Attention: Refer to the Limitations and deviations chapter before using the software for integration.

Intended audience

This document is intended for anyone using the iLLD software.

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1 Release contents

1.1 Release overview

This is Patch Release of iLLD SW for the AURIX™ 3G TC4xx platform.

1.2 Released items

This release contains the release items mentioned in Table 1

Table 1 Release content

Package content	Description
iLLD-TC4-v2.4.1.zip	iLLD package

iLLD-TC4-v2.4.1.zip contains the following items mentioned in Table 2

Table 2 Release zip content

Folder/File	Description
doc	iLLD documentation
examples	iLLD template projects
src	iLLD source for TC4Dx device
iLLD-TC4-v2.4.1-Releasenotes.pdf	A3G iLLD Release Notes

The Hardware functional blocks supported in TC4Dx are mentioned in Table 3

Table 3

Hardware functional block	iLLD driver
CPU	Cpu, Stm
NVM	Flash
SDMA	Dma
PPUC	Ppuc
GETH	Geth
HSPHY	Hsphy
HSSL, HSCT	Hssl
XSPI	Xspi
PCIE	Ep, Rc
DRE	Dre
MCMCAN	Can, Cre
ERAY	Eray
CAN-XL	Canxl
ASCLIN	Asc, Spi, Lin

SENT	Sent
PSI5	Psi5
PSI5s	Psi5s
MSC	Msc
SDMMC	Sd, Emmc
QSPI	QspiMaster, QspiSlave
I2C	I2c
CLOCK	Clock
VMT	Vmt
SCU	Eru
PMS	Pms
SMM	SmmReset, SmmSysMode
IR	Src
WTU	Wtu
SMU	Smu, SmuStdby
FCE	Fce
eGTM	Tom, Atom, Tim, Trigger, Pwm, Input
ADC	Tmadc, Cdsp, Dsadc, Fcc
LETH	Leth
PORTs	Port
AP	Apu, Prot
SCR	AdcComp, Cpu, Port, Rtc, Scu, Ssc, T2ccu, Timers, Uart, Wcan,I2C

1.3 Compatibility

The iLLD software now includes support for C++ compilation. To accommodate this new feature, several modifications have been made to the make options. For users who are utilizing iLLD in C projects and do not wish to alter their make options, it is necessary to incorporate specific C++ dummy functions into the "Ifx_Cfg_Ssw.c" file to enable successful compilation.

```
#ifdef __TASKING__
void _main(void)
{}
void _doexit(void)
{}
#elif defined(__HIGHTEC__) && !defined(__clang__)
void _init(void)
{}
#elif defined(__GNUC__) && !defined(__HIGHTEC__)
void _init(void)
{}
#elif defined(__ghs__)
void _main(void)
{}
void exit(int a)
{}
#endif
```

Add "atexit" macro in Ifx_Cfg_Ssw.h
#define atexit Ifx_atexit

Add below lines of code in Ifx_Cfg_Ssw.c file:

```
#if defined(HIGHTEC) && defined(clang)
void __libc_init_array(void)
{}

void __libc_fini_array (void)
{}

int Ifx_atexit(void (*func)(void))
{
return 0;
}

#endif
```

2 Tool information

The tool information is provided in the following table.

Table 4 Tool details

Tool description	Version details
Compiler	Hightec LLVM v10.0.0 TC CTC SmartCode v10.3r1p1 Metaware_Aurix v2.1/ TC CARC SmartCode v10.3r1p1 GreenHills v2025.1 Windriver v5.9.9.0 GCC 11.3.1 20221230 Built 2025-02-25
Processor platform	TC4Dx
Evaluation hardware	TC4Dx_VDK_v2.3.0 TC4Dx: Target hardware (Triboard) TC4Dx: Target hardware (Emulation)

3 Summary of changes

3.1 Issues fixed in release v2.4.1

This is the first public release for the product, hence no issues are listed.

4 Known Issues

Key	Summary	Impact	Workaround
-	-	-	-

5 Limitations and deviations

This chapter describes the limitations and deviations due to software/hardware design constraints.

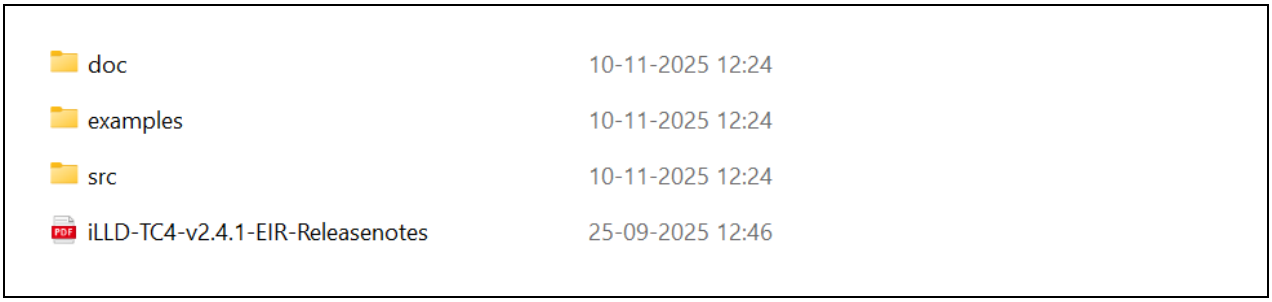
- None

6 Installation

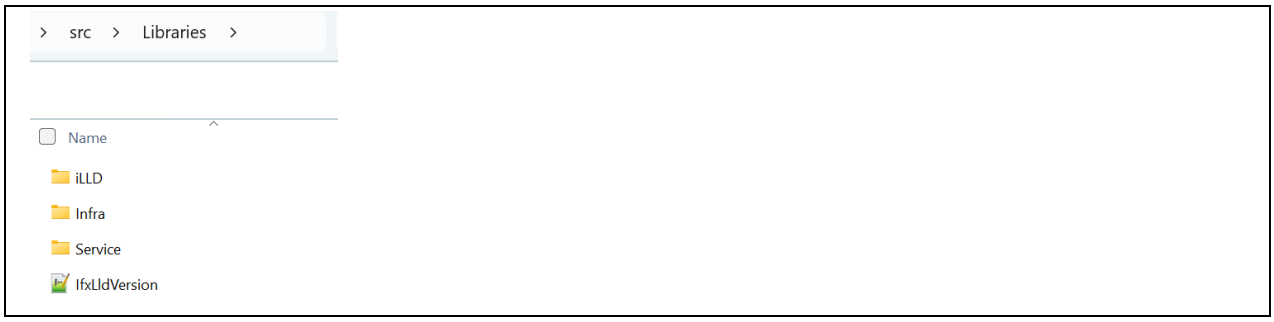
The iLLD software release package is an archive, and it may be extracted to any drive of the computer. There is no specific directory structure needed.

6.1 Package usage

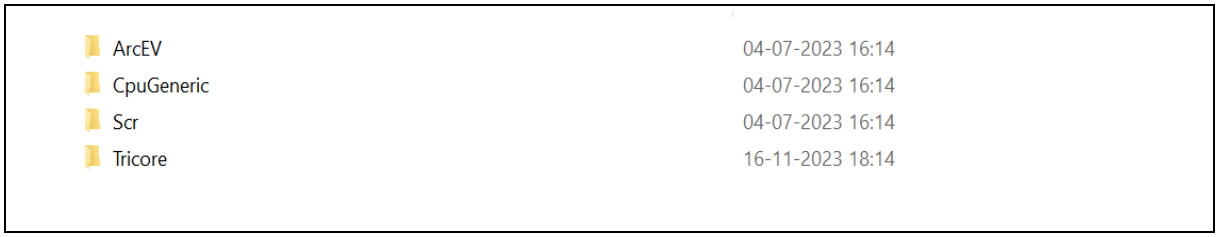
- Copy the iLLD-TC4-<version>.zip file into local drive.
- Package contains the following items.
  - doc : Contains the user manual for the drivers
  - examples : Contains the template project for TC4Dx devices
  - src : Contains Source Code for the TC4Dx drivers
  - iLLD-TC4-<version>-Releasenotes : Aurix3G iLLD Release Note



  - src/Libraries/
    - iLLD : Directory contains the drivers for peripherals of different architecture
    - Infra : Directory contains sfr's, startup software and platform files
    - Service : Directory contains service layer libraries



  - src/Libraries/iLLD/TC4xx/
    - ArcEv : Directory contains drivers for ArcEv
    - Scr : Directory contains drivers for Scr
    - Tricore : Directory contains drivers for Tricore
    - CpuGeneric : Directory contains drivers common for both Tricore and ArcEv



- The iLLD package contains drivers intended for multiple targets(Scr, Tricore, ArcEv). Template projects listed below are provided as part of package.

Target	Template
Tricore	BaseFramework_TC4Dx BaseFramework _TC4Dx_Scr BaseFramework_TC4Dx_Ppu HypervisorDemo_TC4Dx

Guidelines for SCR Baseframework

Within the SCR package, the i51 core drivers are exclusively compiled using the Tasking compiler tool. The Tricore, on the other hand, is compiled using the supported compiler tool that is currently available.

Once the compilation process is completed without errors, the next step involves flashing the Tricore elf. The Tricore will load a C array into the XRAM memory. This array contains all the necessary information required for the SCR boot-up process.

Guidelines for Ppu Baseframework

The PPU drivers are compiled using the MetaWare Compiler. It is essential to verify the presence of predefined libraries for the TC4xx derivatives. If these libraries are included in the current version of MetaWare, you may proceed with the compilation process.

However, if the predefined libraries for the TC4xx derivatives are not available in the installed version of MetaWare on your system, please follow the steps below to install them:

1. Initiate the command prompt by pressing the combination of the Windows key + R.
2. Input the following command: *cmd /V:ON /k C:<MetaWare-path>setup.bat*. This command will activate the Python environment that is bundled with MetaWare.

To download the predefined libraries of a specific derivative, use the following command:

build -ca -sc < derivative>_system -e BUILD_CONFIG=< derivative>_PPU -e BUILD_TYPE=debug

For instance, for the 4Dx derivative, the command would be:

build -ca -sc tc4Dx_system -e BUILD_CONFIG=TC4D_PPU -e BUILD_TYPE=debug

This command will initiate the download of the predefined libraries for the specified derivative.

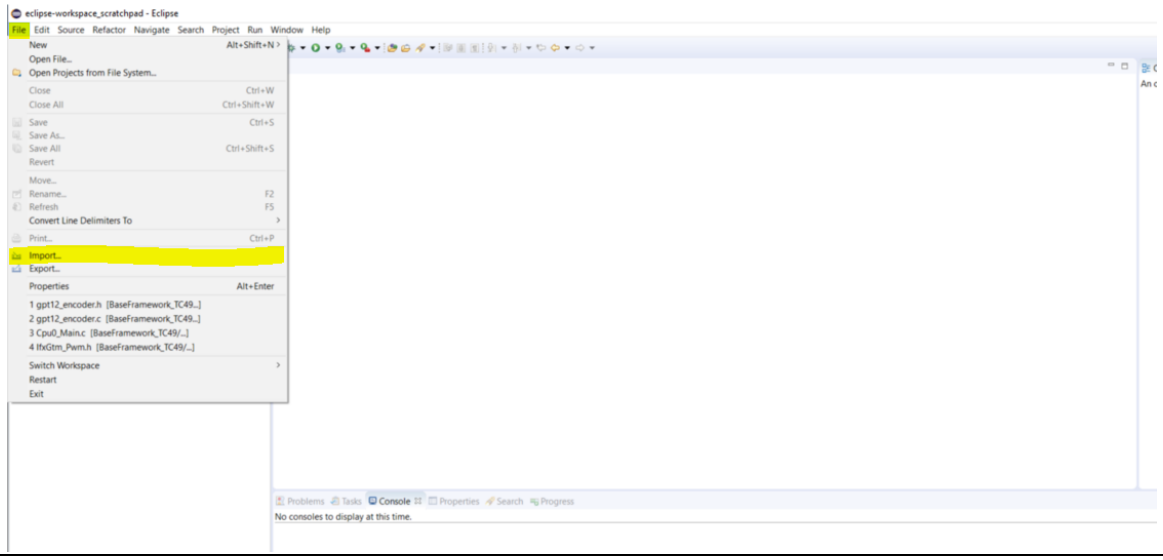
6.2 Importing Baseframework to Bifaces

- Import the template project from the package installation directory

  C:\Infineon\iLLD-TC4-<version>/examples/<project>

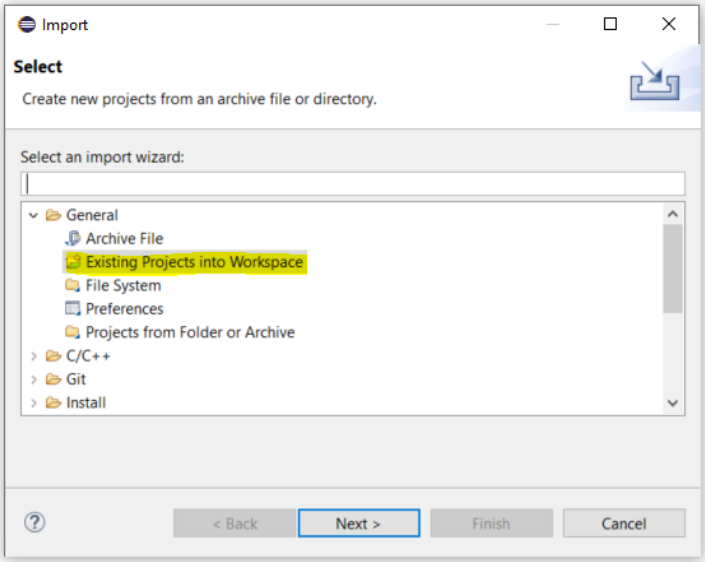
- Open Bifaces. Click on File --> Import

.

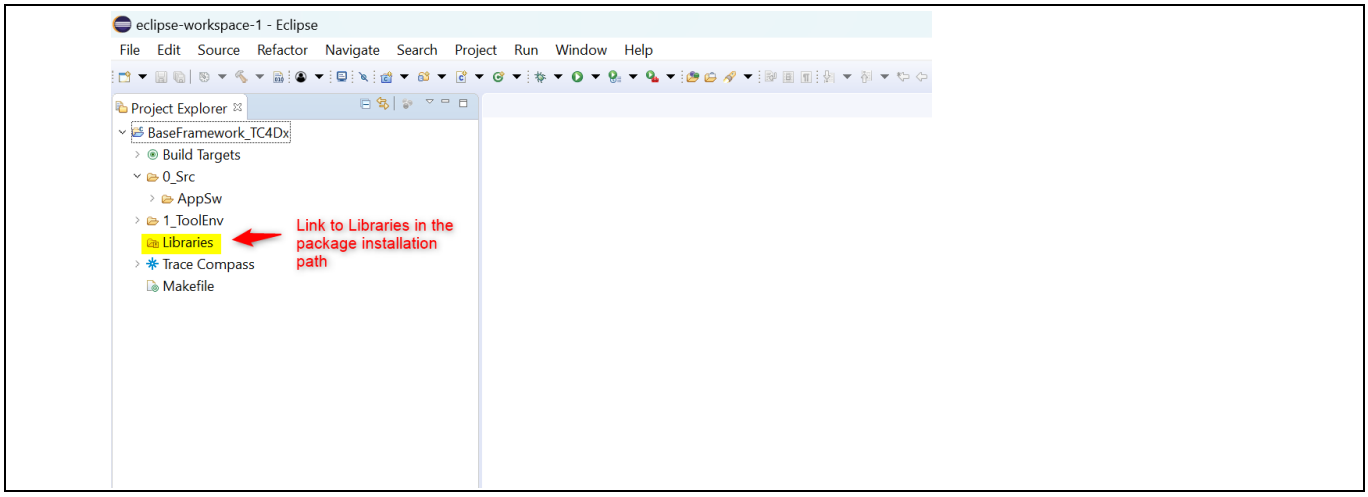


- Select existing project into workspace and Click Next.

.



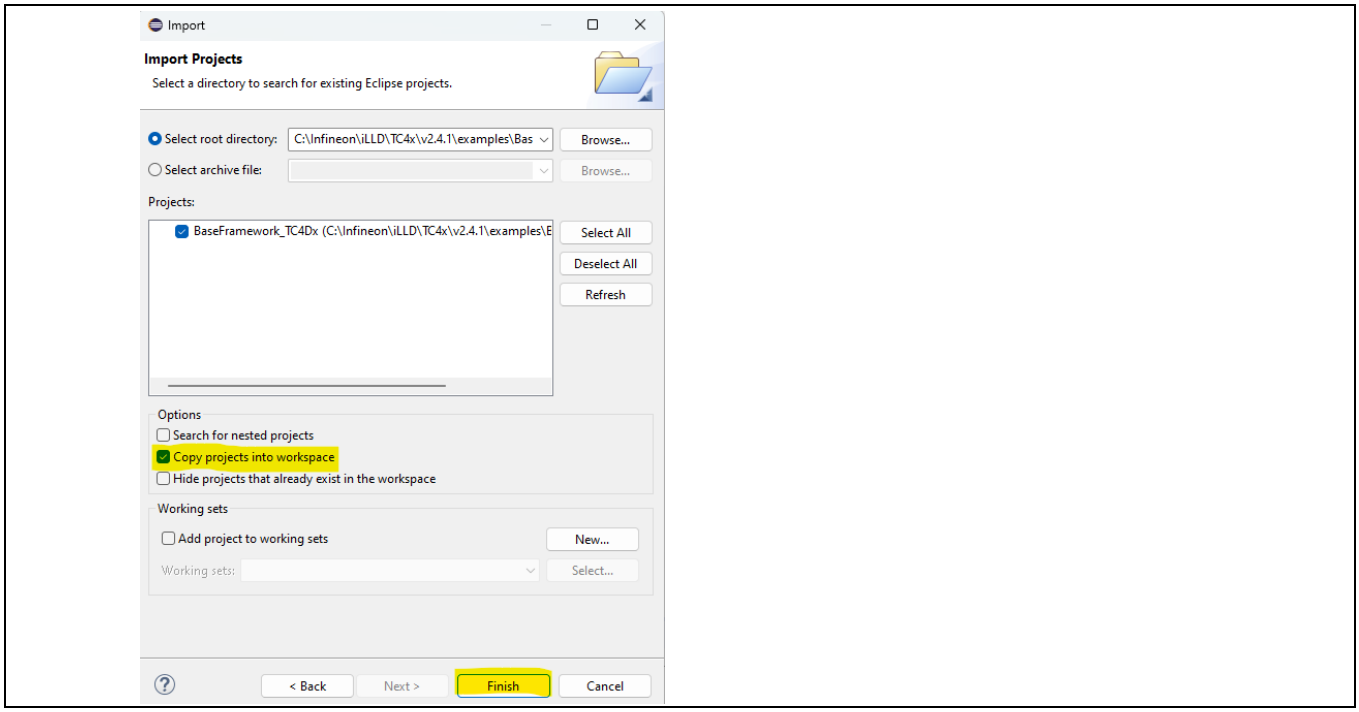
- Click on Browse and select the root directory where the Baseframework is present and Click ok.



Note: The libraries are not included in the template projects, with the exception of the Hypervisor Baseframework. Instead, the libraries are located in the package installation path and are linked to the template project. This means that after importing the template project into BIFACES, users can compile the project as usual, and the required libraries will be automatically linked..

- If you want to copy the Baseframework_TC4Dx into workspace, check the box copy project into workspace. Click Finish and the project will appear in Project Browser.

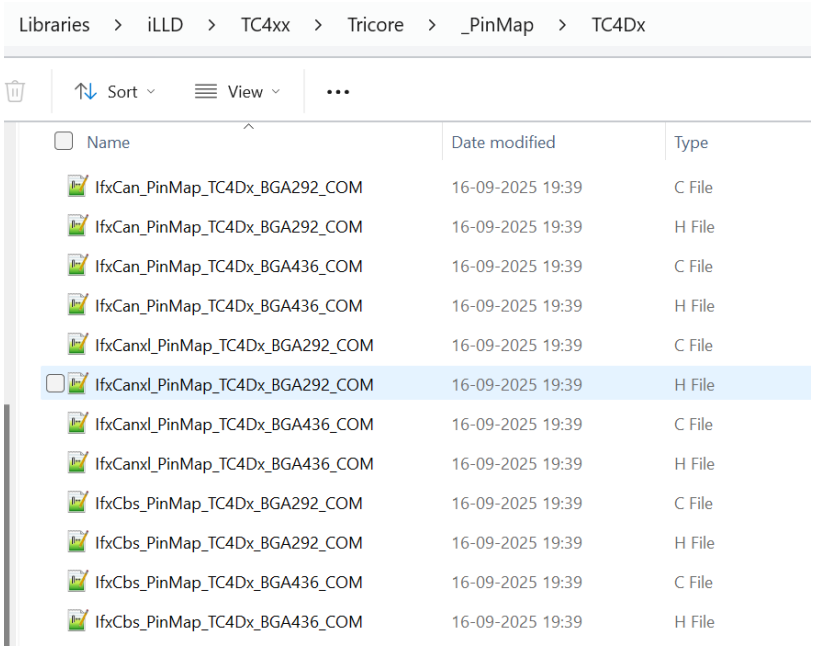
Note: The libraries should also be copied to workspace along with the Baseframework.
.



Note: The iLLD package contains drivers intended for multiple targets (Tricore, SCR, ArcEv). To compile the drivers, please use the BaseFramework provided as reference, or ensure that drivers intended for SCR target are not built for Tricore.

6.3 Selection of Pin package

- PinMap directory contains all the pinmaps for all the packages I.e. there are no separate folders for different packages.



- To select between the different packages, follow the below steps.
    - Packages can be selected with the help of the macros mentioned below, these should be defined in Ifx_Cfg.h
    - For example, in TC4Dx there are two packages available, so define macros as mentioned below
    - To use the BGA292_COM package, set the value of macro IFX_PIN_PACKAGE_BGA292_COM to 1
      #define IFX_PIN_PACKAGE_BGA292_COM 1
    - To use the BGA436_COM package, set the value of macro IFX_PIN_PACKAGE_BGA436_COM to 1
      #define IFX_PIN_PACKAGE_BGA436_COM 1

Note: Only one macro should be defined in Ifx_cfg.h as explained in below screenshot.

File: Ifx_Cfg.h

```
/* Change Package by defining macro */
#define IFX_PIN_PACKAGE_BGA292_COM 1
// #define IFX_PIN_PACKAGE_BGA436_COM 1
```

6.4 Compiler Selection to build iLLD

- Multiple compiler toolchains (mentioned in Table Tool information) are supported to build Tricore iLLD's
- Compiler selection is handled in Config.xml present in '**1_ToolEnv\0_Build\1_Config\Config.xml**'

.

```
<!-- Architecture congfigurations -->
<architecture name="Tricore" primaryToolchain= "Gnuc">
    <!-- Target congfigurations -->
    <target name="Tc" enable="true">
        <output type="elf" enable="true" fileName="$(PROJ_NAME)_Tc.elf" tool="Ld"/>
        <output type="hex" enable="true" fileName="$(PROJ_NAME)_Tc.hex" tool="Hex">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="srec" enable="false" fileName="$(PROJ_NAME)_Tc.srec" tool="Srec">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="lib" enable="false" fileName="$(PROJ_NAME)_Tc.a" tool="Ar"></output>
    </target>
```

6.5 Compiler Path to build iLLD

- Compile path is specified in '1_ToolEnv\0_Build\1_Config\Config_Tricore_{compiler}\Config_{compiler}.mk'

```
1_ToolEnv
  0_Build
    1_Config
      Config_ArcEV_Metaware
      Config_Tricore_Ghs
      Config_Tricore_Gnuc
        Config_Gnuc.mk
        Lcf_Gnuc_Tricore_Tc.lsl
      Config_Tricore_Hightec
      Config_Tricore_Tasking
    BifacesConfig.xsd
    Config.mk
    Config.xml
```

```
16 # MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE APPLY TO THIS SOFTWARE.#
17 # INFINEON SHALL NOT, IN ANY CIRCUMSTANCES, BE LIABLE FOR SPECIAL, INCIDENTAL,#
18 # OR CONSEQUENTIAL DAMAGES, FOR ANY REASON WHATSOEVER.                         #
19 #                                                                              #
20 ################################################################################
21
22 B_GNUC_TRICORE_PATH:= C:\Tools\HIGHTEC\gnutri_2020.03-b3513f1-win64
23
24
25 B_GNUC_TRICORE_CC_OPTIONS= -march=tc18 -mcpu=tc4xxx -g -O2 -fno-common -fstrict-volatile-bitfields \
26                            -ffunction-sections -fdata-sections -Wall -std=c11
27
28
29 B_GNUC_TRICORE_ASM_OPTIONS= $(GNUC_TC_CC_OPTIONS)
```

7 Support packages

No Support packages are provided in this release

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